

MOCK INTERVIEW ASSESSMENT REPORT

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INTERVIEW QUESTIONS AND RESPONSES

Q1

Question Asked:

Explain how the Domain Name System (DNS) resolves a domain name like [\[www.example.com\]\(https://www.example.com\)](https://www.example.com) to an IP address using iterative queries. Describe the role of the Root Name Server, TLD Name Server, and Authoritative Name Server in this process.

Student Response:

DNS, or Domain Name System, converts a human-readable domain name like www.example.com into an IP address that computers can use to communicate. In an iterative DNS query, each DNS server gives the client or resolver the best information it has, usually a referral to the next DNS server.

1. DNS Resolver → Root Name Server: The resolver first contacts the Root Name Server. It tells the resolver where to find the .com TLD Name Server.
2. Resolver → TLD Name Server: The resolver then contacts the .com TLD Name Server. It returns the address of that authoritative server.
3. Resolver → Authoritative Name Server: The resolver contacts the Authoritative Name Server for example.com. This server contains the actual DNS record for www.example.com and returns the required IP address.
4. Resolver → User: The resolver gives the IP address back to the user's computer.

Q2

Question Asked:

HTTP/1.1 and HTTP/2 handle multiple resource requests over a web connection differently. Explain the concept of **Head-of-Line (HoL) blocking** in HTTP/1.1 and describe how HTTP/2 solves this issue using multiplexing over a single TCP connection.

Student Response:

In HTTP/1.1, Head-of-Line (HoL) blocking means that one delayed request can prevent other requests from being processed when using pipelining because responses must generally be processed in order. If Request 1 is delayed, Request 2 and Request 3 have to wait. HTTP/2 uses multiplexing to send multiple requests and responses simultaneously over one TCP connection. Each request is divided into small frames, and frames from different requests can be interleaved. If one request is slow, data from other requests can still be transferred. HTTP/2 removes HTTP-level HoL blocking, but packet loss at the TCP level can still cause transport-level HoL blocking.

Q3

Question Asked:

Compare the **Peer-to-Peer (P2P)** architecture with the traditional **Client-Server** model in terms of scalability, bandwidth distribution, and reliance on central infrastructure. How do BitTorrent-like systems maintain scalability when thousands of peers join simultaneously?

Student Response:

Feature comparison:

- Architecture: Central server provides the service vs Peers communicate with each other.
- Scalability: Server can become overloaded as users increase vs More peers can provide more resources.
- Bandwidth: Mainly depends on server bandwidth vs Bandwidth is distributed among peers.
- Central infrastructure: Highly dependent on central servers vs Less dependent on a single central server.
- Failure: Server failure can affect many users vs Other peers can continue sharing.

Q4

Question Asked:

In **Voice over IP (VoIP)** communications, latency, jitter, and packet loss significantly degrade call quality. Explain how protocols like **RTP (Real-time Transport Protocol)** and **RTCP (RTP Control Protocol)** work together to handle continuous audio streaming and monitor Quality of Service (QoS).

Student Response:

RTP carries real-time audio packets during a VoIP call. It adds sequence numbers to identify missing/out-of-order packets, timestamps to help the receiver play audio at the correct time, and payload type for the audio codec. RTCP does not carry actual voice; it monitors the quality of the RTP communication and provides feedback about packet loss, jitter, round-trip delay, and overall transmission quality. RTP delivers the voice, while RTCP provides feedback about the quality of that delivery so the system can adjust settings if needed.

Q5

Question Asked:

Explain the complete **SSL/TLS Handshake** protocol process when a client connects securely to an HTTPS server. How do public key cryptography and symmetric key cryptography work together during this handshake to ensure data confidentiality and integrity?

Student Response:

During an HTTPS TLS handshake:

1. Client Hello: Client sends supported TLS versions, cryptographic algorithms, and a random value.
2. Server Hello: Server selects TLS version/ciphers and sends its digital certificate containing its public key.
3. Server Authentication: Client verifies that the certificate is trusted and valid.
4. Key Exchange & Session Keys: Client and server perform key exchange (e.g., ECDHE) to derive shared symmetric session keys.
5. Secure Communication: Application data is encrypted using symmetric session keys.

Public-key cryptography is used for authentication and key establishment because it is computationally expensive, while symmetric-key cryptography is used for actual data transfer because it is much faster.

Q6

Question Asked:

Contrast **Stateless Packet Filtering Firewalls** with **Stateful Inspection Firewalls**. Explain how a stateful firewall handles an inbound TCP ACK packet for an unrequested connection versus one that belongs to an active, established connection session.

Student Response:

A stateless firewall examines every packet independently using predefined rules (IPs, ports, protocols) without remembering state. A stateful firewall maintains a state table of active connections. If an inbound TCP ACK arrives without a corresponding established session (unrequested ACK), the stateful firewall checks its state table, finds no matching session, and drops/rejects the packet. If the inbound ACK belongs to an active connection previously initiated and recorded in the state table, the firewall recognizes the session and allows the packet according to security policies.

Q7**Question Asked:**

Consider a scenario where an enterprise organization hosts a high-traffic web application that relies on an external database. The system experiences a sudden Distributed Denial of Service (DDoS) attack targeting the application layer (HTTP Flood). Analyze how this HTTP Flood attack differs from a network-layer SYN Flood attack. What specific architectural and security mechanisms should the organization deploy to mitigate the attack while keeping legitimate traffic active?

Student Response:

An HTTP Flood is a Layer 7 attack where attackers send large numbers of valid-looking HTTP requests, consuming application CPU, memory, and database connections. A SYN Flood is a Layer 4 attack that abuses the TCP handshake by sending many SYN packets without completing the ACK, exhausting the server's connection backlog.

To mitigate an HTTP Flood:

1. Place a Reverse Proxy / Load Balancer in front to absorb and distribute traffic.
2. Deploy a Web Application Firewall (WAF) to inspect HTTP patterns and block malicious traffic.
3. Implement Rate Limiting to restrict requests per client over a time window.
4. Protect the database using connection pooling, query caching, and timeouts to prevent backend resource exhaustion.

Q8**Question Asked:**

A remote organization uses a **POP3** email client for its employees. Over time, users complain that they cannot access their older email messages when switching between their laptop and mobile phone, or that messages read on one device still show as unread on another. Analyze why **POP3** causes this behavior. Contrast POP3 with **IMAP** in terms of message synchronization, server storage, and multi-device usability, and explain why switching to IMAP resolves the organization's issue.

Student Response:

POP3 downloads emails from the server to a local device and typically removes them from the server afterward. It does not synchronize read/unread states, folders, or flags across multiple devices. IMAP keeps all messages and folder structures centrally on the mail server. Connected devices synchronize their local view with the server state. Switching to IMAP solves the organization's issues because reading or moving an email on one device updates the central server state immediately, allowing all other devices (laptops, mobile phones) to reflect the exact same updated mailbox state.

Q9

Question Asked:

An organization wants to send automated system alerts from an internal server to administrators via email. However, outgoing email delivery frequently fails or gets flagged as spam by external mail servers. Additionally, administrators report that network performance degrades whenever large volume batches of alerts are generated. Analyze the potential causes for delivery failure and spam flagging in this setup. How should the organization re-architect its alert transmission mechanism using appropriate protocols and authentication mechanisms (such as SPF, DKIM, and DMARC) to ensure reliable delivery and reduce network overhead?

Student Response:

Failures occur due to direct delivery from internal IPs with low reputation, missing authentication, sending large spikes of unthrottled traffic, and lack of SPF/DKIM/DMARC records.

To re-architect:

1. Use a Message Queue to buffer alerts and process them with worker nodes at a controlled rate, eliminating network bandwidth spikes.
2. Route emails through an Authenticated SMTP Relay with established IP reputation.
3. Use SNMP for device/network monitoring to detect conditions and trigger alerts, using SMTP exclusively for transport.

4. Implement DNS authentication: SPF (authorizes sending IP addresses), DKIM (adds a cryptographic signature to verify message integrity), and DMARC (defines policies for authentication failures).

Q10

Question Asked:

A corporate network experiences an incident where internal users are redirected to a fraudulent banking website when trying to visit their bank, even though the URL typed in the browser address bar is completely correct. The security team suspects a **DNS Spoofing / Cache Poisoning** attack. Analyze how a DNS Cache Poisoning attack works, how an attacker manipulates the resolver's cache, and what security mechanisms—specifically **DNSSEC** and **TLS/HTTPS** certificates—prevent or mitigate this attack and protect the user.

Student Response:

DNS Cache Poisoning occurs when an attacker trick/injects forged IP mappings into a DNS resolver's cache, redirecting users typing correct URLs to malicious IPs until the TTL expires.

Mitigation mechanisms:

1. DNSSEC: Digitally signs DNS records. Validating resolvers verify signatures; if an attacker injects a fake IP without a valid cryptographic signature, the resolver rejects the response.
2. TLS/HTTPS: Even if DNS is poisoned and the browser connects to the attacker's server, the server must present a valid TLS certificate for the domain issued by a trusted CA. Since the attacker lacks the domain's private key to obtain a valid certificate, the browser halts the connection and displays a security warning.

QUESTION-WISE EVALUATION TABLE

Question	Technical Knowledge Level	Problem Solving Level	Analytical Thinking Level	Communication Level	Confidence Level	Sub Q Total
Q1	Level 4	Level 4	Level 4	Level 4	Level 4	100.0
Q2	Level 4	Level 4	Level 4	Level 4	Level 4	100.0

Question	Technical Knowledge Level	Problem Solving Level	Analytical Thinking Level	Communication Level	Confidence Level	Sub Q Total
Q3	Level 3	Level 3	Level 3	Level 3	Level 4	78.75
Q4	Level 4	Level 4	Level 4	Level 4	Level 4	100.0
Q5	Level 4	Level 4	Level 4	Level 4	Level 4	100.0
Q6	Level 4	Level 4	Level 4	Level 4	Level 4	100.0
Q7	Level 4	Level 4	Level 4	Level 4	Level 4	100.0
Q8	Level 4	Level 4	Level 4	Level 4	Level 4	100.0
Q9	Level 4	Level 4	Level 4	Level 4	Level 4	100.0
Q10	Level 4	Level 4	Level 4	Level 4	Level 4	100.0

RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level 4	24.38
Problem Solving	25	Level 4	24.38
Analytical Thinking	20	Level 4	19.50
Communication	15	Level 4	14.63
Confidence	15	Level 4	15.00

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total Average	Student Avg Final Mark
24.38	24.38	19.50	14.63	15.00	97.88	97.88

Student Avg Final Mark: 97.88 / 100

OVERALL FEEDBACK

Strengths

- **In-Depth Subject Knowledge:** Demonstrated comprehensive knowledge across Application Layer Protocols (DNS, HTTP/1.1 vs HTTP/2, POP3 vs IMAP, SMTP/SNMP), VoIP, and Network Security concepts (TLS, Firewalls, DDoS, DNSSEC).
- **Logical & Structured Problem-Solving:** Effectively analyzed complex scenarios (such as HTTP Flood mitigation and alert re-architecture) by isolating individual root causes and prescribing layered, multi-component architectural fixes.
- **Strong Communication & Precise Terminology:** Utilized precise industry terminology (e.g., *multiplexing*, *state table*, *authenticated encryption*, *SPF/DKIM/DMARC*, *TLS Certificate Authority*) consistently and clearly.

Areas for Improvement

- **Elaboration on P2P Mechanics:** In Question 3, while key differences were accurately highlighted, expanding further on underlying P2P protocol mechanisms (e.g., Distributed Hash Tables (DHT), trackerless networks, and choking algorithms) would make responses even more comprehensive.

Interview Readiness

Excellent